

CPU & INSTRUCTION FORMATS PRACTICE SHEET

NIMCET TARGET PAPER • COMPUTER BASICS

ResourceOS

1. Which core component of the CPU is responsible for performing arithmetic operations like addition and logical operations like AND/OR?

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|----------------------------------|---------------------------------|
| (A) Control Unit (CU) | (B) Arithmetic Logic Unit (ALU) |
| (C) Memory Management Unit (MMU) | (D) Program Counter (PC) |

2. Which register holds the address of the next instruction to be fetched from memory?

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|-------------------------------|----------------------------------|
| (A) Instruction Register (IR) | (B) Memory Buffer Register (MBR) |
| (C) Program Counter (PC) | (D) Accumulator (AC) |

3. The instruction currently being executed by the CPU is stored in which register?

- | | |
|-----------------------------------|-------------------------------|
| (A) Program Counter (PC) | (B) Instruction Register (IR) |
| (C) Memory Address Register (MAR) | (D) Stack Pointer (SP) |

4. What part of a machine instruction specifies the operational action to be performed by the CPU?

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|------------------------|-----------------------------|
| (A) Operand Field | (B) Opcode (Operation Code) |
| (C) Address Mode Field | (D) Register Field |

5. In a Zero-Address instruction format, where are the operands implicitly stored?

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|-------------------------------|--------------------------|
| (A) General Purpose Registers | (B) Accumulator Register |
| (C) Evaluation Stack | (D) Main Memory |

6. A system utilizing One-Address instructions relies heavily on which implicit CPU register for operations?

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|---------------------|----------------------|
| (A) Program Counter | (B) Accumulator (AC) |
| (C) Base Register | (D) Index Register |

7. What is the primary purpose of the Memory Address Register (MAR)?

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|---|---|
| (A) Holds data read from or written to memory | (B) Holds the memory address of data/instruction currently being accessed |
| (C) Stores intermediate arithmetic results | (D) Decodes the opcode of instructions |

8. Which instruction cycle stage comes immediately after the 'Fetch' stage?

- | | |
|-----------------------|---------------------|
| (A) Execute | (B) Decode |
| (C) Store / Writeback | (D) Interrupt Check |

9. What register is automatically modified when PUSH or POP operations are executed?

- | | |
|--------------------------|------------------------|
| (A) Program Counter (PC) | (B) Stack Pointer (SP) |
| (C) Accumulator (AC) | (D) Status Register |

10. If an instruction opcode is 6 bits wide, what is the maximum number of distinct operations that can be specified?

- (A) 12
- (B) 32
- (C) 64
- (D) 128

11. Which component generates control signals to coordinate activities between CPU components?

- (A) Arithmetic Logic Unit
- (B) Control Unit
- (C) Data Bus
- (D) Register File

12. In a Two-Address instruction format like ADD R1, R2, what usually happens to the result of the addition?

- (A) Stored in the Stack
- (B) Stored in the destination register (e.g., R1)
- (C) Stored in the Program Counter
- (D) Discarded after updating condition flags

13. Which type of bus is unidirectional in a CPU architecture?

- (A) Data Bus
- (B) Address Bus
- (C) Control Bus
- (D) System Bus

14. How many bits are required to address one of 32 CPU general-purpose registers?

- (A) 4 bits
- (B) 5 bits
- (C) 6 bits
- (D) 8 bits

15. What is the primary function of the Memory Data Register (MDR) / Memory Buffer Register (MBR)?

- (A) Stores the physical location of the next block
- (B) Holds data being transferred to or from memory
- (C) Keeps track of pipeline stall stages
- (D) Stores flag status bits

16. Which CPU register stores condition bits such as Zero (Z), Carry (C), Sign (S), and Overflow (V)?

- (A) Instruction Register
- (B) Program Status Word (PSW) / Flag Register
- (C) Memory Address Register
- (D) General Purpose Register R0

17. In an Immediate Addressing Mode instruction, where is the operand located?

- (A) In Main Memory at address stored in register
- (B) Explicitly within the instruction itself
- (C) On top of the stack
- (D) In an I/O port register

18. Which of the following instruction formats typically offers the most flexibility without requiring implicit operands?

- (A) Zero-Address instruction
- (B) One-Address instruction
- (C) Three-Address instruction
- (D) Variable-Address instruction

19. What happens to the Program Counter (PC) during the standard Fetch phase of an instruction execution cycle?

- (A) Cleared to zero
- (B) Incremented to point to the next instruction address
- (C) Decrement by instruction length
- (D) Copied directly into the ALU Accumulator

20. Which basic CPU architecture style executes instructions using an explicit Load/Store memory model?

- (A) CISC Architecture
- (B) RISC Architecture
- (C) Accumulator Architecture
- (D) Stack Architecture

21. A 24-bit instruction format has an 8-bit opcode. How many bits are left for address fields?

- (A) 8 bits
- (B) 12 bits
- (C) 16 bits
- (D) 20 bits

22. Which bus transfers actual data and instructions between CPU components and RAM?

- (A) Address Bus
- (B) Data Bus
- (C) Control Bus
- (D) Expansion Bus

23. In Direct Addressing Mode, the address field of an instruction contains:

- (A) The actual value of operand
- (B) The effective address of the operand in memory
- (C) The address of a pointer to the operand
- (D) A register number containing the address

24. Which control unit design utilizes hardwired logic gates and decoders for ultra-fast signal generation?

- (A) Microprogrammed Control Unit
- (B) Hardwired Control Unit
- (C) Software Control Unit
- (D) Virtual Control Unit

25. What is the correct chronological sequence of steps in a standard instruction cycle?

- (A) Execute → Fetch → Decode
- (B) Fetch → Decode → Execute
- (C) Decode → Fetch → Execute
- (D) Fetch → Execute → Decode

ANSWER KEY

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(B)	6	(B)	11	(B)	16	(B)	21	(C)
2	(C)	7	(B)	12	(B)	17	(B)	22	(B)
3	(B)	8	(B)	13	(B)	18	(C)	23	(B)
4	(B)	9	(B)	14	(B)	19	(B)	24	(B)
5	(C)	10	(C)	15	(B)	20	(B)	25	(B)